Cyclistic Data Cleaning Log

Project: Google Data Analytics Capstone – Cyclistic Case Study

This document records the data preparation and cleaning steps performed prior to analysis in PostgreSQL and Power BI.

Step	Action	Reason / Notes
1	Collected all CSV files from 2020 onwards into a single folder.	Prepared the monthly datasets for consolidation.
2	Combined all CSV files into a single dataset using Python (pandas).	Created one analysis-ready dataset from the monthly files.
3	Uploaded the combined dataset into PostgreSQL.	Enabled SQL-based cleaning and analysis.
4	Replaced NULL values in start_station_name and start_station_id with 'Unknown'.	Retained records while clearly identifying missing station information.
5	Created a working copy of the table.	Preserved the original imported table before further cleaning.
6	Removed duplicate ride_id records from the working copy.	Ensured each ride was represented only once.
7	Replaced NULL values in end_station_name and end_station_id with 'Unknown'.	Retained records while identifying missing destination station information.
8	Converted started_at and ended_at to TIMESTAMP data types (started_at_ts and ended_at_ts).	Enabled date/time calculations.
9	Created ride_duration (INTERVAL) and ride_duration_minutes columns.	Prepared the dataset for ride duration analysis.
10	Created analytical columns: hour, end_hour, month and day of week (DOW).	Prepared the data for time-based analysis and reporting.
11	Removed rides with negative durations.	Ride end times occurring before start times were treated as invalid records.
12	Removed rides lasting longer than 24 hours.	Represented approximately 0.1% of the dataset and were treated as extreme outliers.
13	Removed rides shorter than 1 minute.	Represented approximately 2% of the dataset and were

		treated as likely accidental unlocks, test rides or incomplete trips.
14	Removed 11 records where end_lat = 0.	A latitude of 0 is geographically impossible for the Chicago-area bike-share dataset and was treated as a data quality issue.
15	Removed 41 records with implausible geographic coordinates.	Coordinates fell well outside the plausible operating region and were treated as data quality anomalies.
16	Removed the remaining 7 records with suspicious longitude values and missing destination station information ('Unknown').	Multiple indicators suggested incomplete or erroneous records.
17	Standardised inconsistent station-name capitalisation discovered during station analysis.	Corrected the identified 'Lasalle St & Washington St' / 'LaSalle St & Washington St' variation so the same station was represented consistently.

Post-cleaning reporting preparation

The following steps were performed after the core cleaning process to prepare the data for analysis and Power BI reporting. They are included here for completeness but are not treated as additional data-quality cleaning decisions.

Step	Action	Reason / Notes
A	Created the cyclistic_analysis reporting view.	Restricted the Power BI reporting dataset to rides beginning in 2024 onward as the PC in use could not handle more data without running very slowly
B	Created the start_stations station-level table using average latitude and longitude for each start_station_name.	Provided one representative coordinate per station for the Power BI station map and avoided plotting every individual ride coordinate.

Summary

The final dataset contains unique ride records, validated timestamps, cleaned ride durations, reasonable geographic coordinates, completed handling of missing station information, standardised station naming where inconsistencies were identified, and derived date/time fields suitable for SQL analysis and Power BI reporting. A separate SQL workflow file contains the technical implementation of these steps and the subsequent analysis queries.